

Calibrate the method before you trust the method

Five defensible ways to estimate a heat-pump retrofit's saving, each resting on different assumptions. Which of them actually holds its size on the panel this programme will really have?

Abstract

Five defensible ways to estimate a heat-pump retrofit's saving, each resting on different assumptions. Which of them actually holds its size on the panel this programme will really have? Describe the panel you will actually have, not a generic one. 5 steps are recorded in full below. All 5 candidates calibrated: on a panel of this shape their nominal 5% tests really do reject about 5% of the time, so their intervals mean what they say. That is a result, and it is worth having explicitly rather than assuming — a method whose size were off here would poison the power calculation and the business case with it. No assumptions were recorded for this run, which is not the same as none being required.

1. Introduction

This report addresses one question: Five defensible ways to estimate a heat-pump retrofit's saving, each resting on different assumptions. Which of them actually holds its size on the panel this programme will really have?

2. Methods

Question: Five defensible ways to estimate a heat-pump retrofit's saving, each resting on different assumptions. Which of them actually holds its size on the panel this programme will really have?

Describe the panel you will actually have, not a generic one

Every claim about a method's behaviour is conditional on the data it runs on. 120 buildings, 24 months, a year of pre-period, half treated, buildings that differ a lot at baseline, weather that moves everyone together, and month-to-month autocorrelation of 0.6 — those are the features that decide whether a variance estimator is honest here, and none of them are in a textbook's assumptions.

Considered instead: Choosing a method from what the literature reports. The literature reports behaviour on the panels the authors had. A method that holds its size on 500 units with no autocorrelation may not hold it on 120 with $\rho = 0.6$, and nothing about the method's name tells you which case you are in.

```
buildings : 120 (60 treated)
months : 24, of which 12 pre-period
baseline spread : unit sd 1.2, noise sd 0.9
common shocks : period sd 0.6 (weather)
autocorrelation : rho 0.6
simulations each : 120
```

Read what each candidate asks you to believe

The registry carries each method's requirements and its named assumptions, so the comparison starts from what is being assumed rather than from what is familiar. Two methods that produce similar point estimates can rest on completely different claims, and the difference only shows up when one of those claims is false.

Considered instead: Comparing methods on their estimates. On real data they will differ and there is no way to tell which is right; on simulated data with a known truth they will mostly agree, which tells you nothing about the case where it matters.

method	needs	assumes
difference_in_differences	pre-period, controls	parallel_trends, no_anticipation, no_interference
synthetic_control	pre-period, controls	convex_hull, no_anticipation, no_interference
time_based_regression	pre-period, controls	stable_pre_relationship, no_anticipation, no_interference
cluster_based_regression	pre-period, controls	random_assignment, no_interference
switchback	nothing but the assignment	random_assignment, no_carryover, stationary_noise
ghost	controls	random_assignment, exposure_known, no_interference

Table 1

Drop one candidate by hand, and say why

Synthetic control builds a donor combination for a single treated unit. This programme treats half the estate at once, so the method is not wrong — it is answering a different shape of question, and its solver says so by

failing rather than by returning a number. Choosing the candidate set is part of the design.

Considered instead: Looping over everything in the registry and taking whatever survives. That outsources a design decision to an exception handler, and on a different panel it would silently include a method whose question nobody checked.

```
in the registry : ['cluster_based_regression',  
'difference_in_differences', 'ghost', 'switchback', 'synthetic_control',  
'time_based_regression']  
carried forward : ['cluster_based_regression',  
'difference_in_differences', 'ghost', 'switchback',  
'time_based_regression']
```

Calibrate on panels with no effect at all

Run each candidate many times on data generated with a true effect of exactly zero, and count how often it declares one. A nominal 5% test should reject about 5% of the time. If it rejects 15%, every interval it produces is too narrow, and every downstream number — the power calculation, the business case — inherits that.

Considered instead: Trusting the nominal level. It is nominal: it is what the test would deliver under assumptions this panel may not satisfy. Autocorrelation of 0.6 is exactly the kind of thing that breaks a variance estimator quietly.

difference_in_differences 5/120 = 4.2% acceptable [0, 15] stable
time_based_regression 7/120 = 5.8% acceptable [0, 15] stable
cluster_based_regression 4/120 = 3.3% acceptable [0, 15] stable
switchback 11/120 = 9.2% acceptable [0, 15] stable
ghost 4/120 = 3.3% acceptable [0, 15] stable

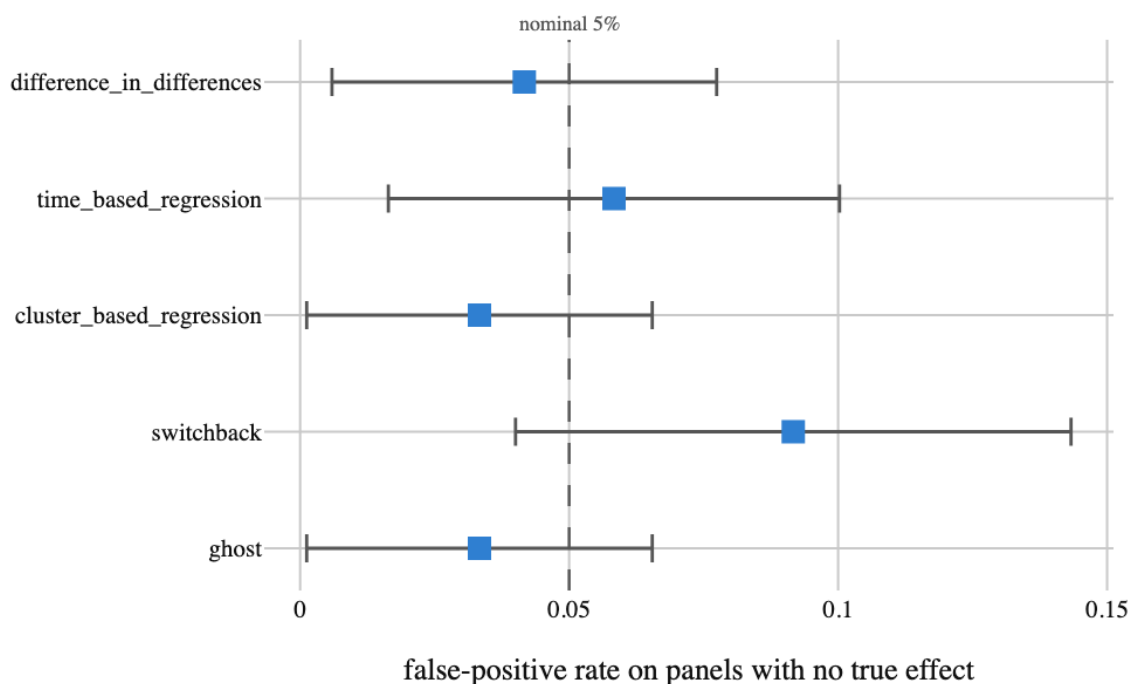


Figure 1. Does a 5% test reject 5% of the time on this panel? Each dot is the observed false-positive rate over the simulations, with a binomial interval. An interval that misses the dashed line is a method whose stated uncertainty is wrong here — and would be drawn in red

Only now look at power, and only for the survivors

Calibration first, power second. The order is the point: a method that cannot hold its size will look powerful precisely because it rejects too often, so ranking on power first selects for the broken one. With size settled, power is a fair comparison and it separates the candidates hard.

Considered instead: Ranking every method by power and taking the top one. That is how a method that cannot hold its size ends up in a signed analysis plan, and by then the false-positive rate is somebody else's problem.

```
difference_in_differences power 1.000 bias -0.004 coverage 0.96
time_based_regression power 0.967 bias +0.001 coverage 0.94
cluster_based_regression power 1.000 bias -0.003 coverage 0.97
switchback power 1.000 bias -0.005 coverage 0.91
ghost power 0.192 bias +0.005 coverage 0.97
```

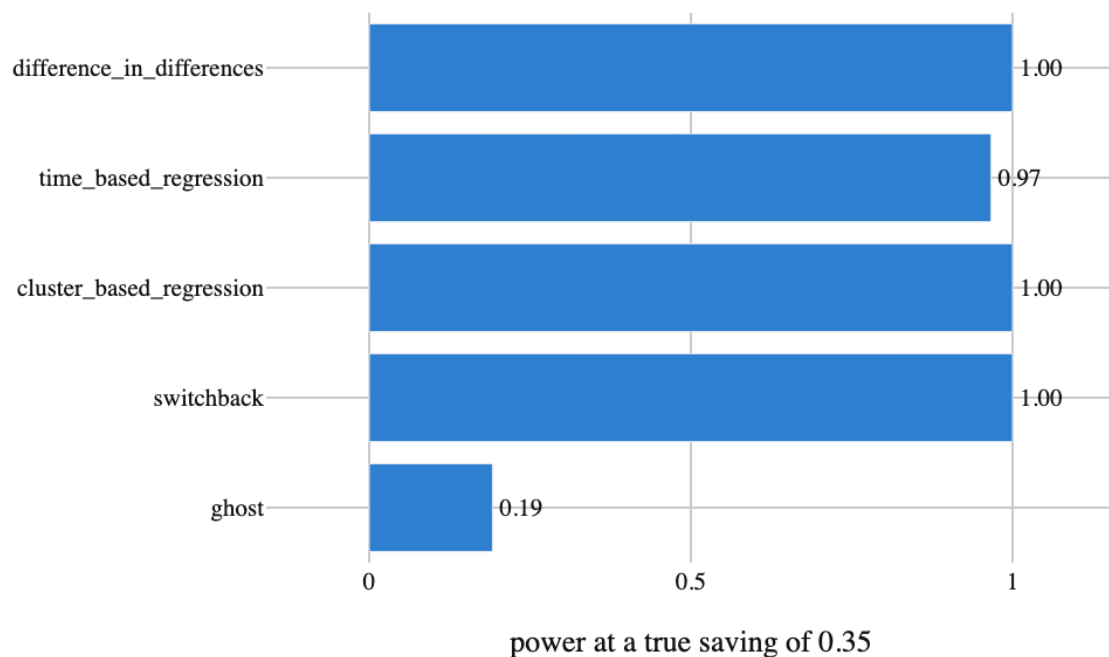


Figure 2. With size settled, power separates them. 'ghost' is not broken. It estimates a different quantity — intent-to-treat among the exposed — from a much smaller effective sample, and pays for it in power. A ranking that had not calibrated first could not have told that apart from a method that simply rejects too often

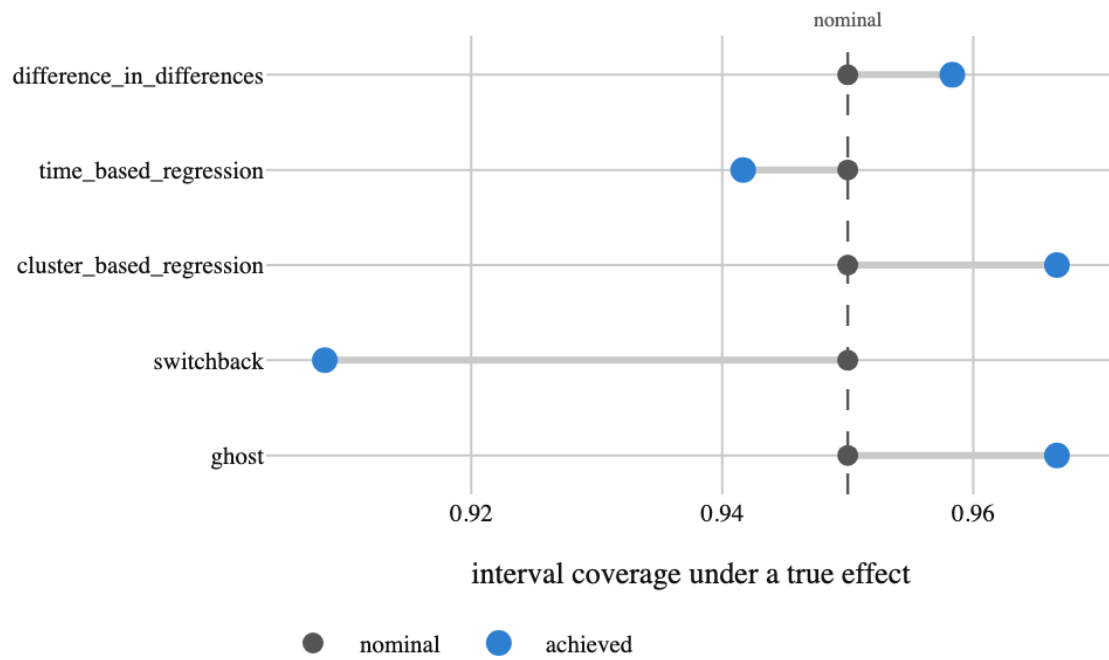


Figure 3. And coverage catches what the size test alone would miss. Size was checked under the null; this is coverage under a real effect. The bar of each dumbbell is the gap between what the method promises and what it delivers. Anything short of nominal is producing intervals that are too narrow when it matters — the kind of detail worth seeing before the analysis plan is signed rather than after

3. Results

No findings were recorded.

4. What the run showed

Written by the analyst after seeing the output, and reproduced unchanged. Nothing in this section is generated.

All 5 candidates calibrated: on a panel of this shape their nominal 5% tests really do reject about 5% of the time, so their intervals mean what they say. That is a result, and it is worth having explicitly rather than assuming — a method whose size were off here would poison the power calculation and the business case with it.

With calibration settled, power separated them hard, and the separation is interpretable rather than mysterious: the low-power method is estimating a different quantity from a smaller effective sample, not failing at the same one.

Worth not skipping past: switchback's coverage runs a little under nominal even though its size test passed. The HAC variance is working against real autocorrelation here, and the shortfall only shows up once there is an effect to cover. A size check under the null is necessary and it is not sufficient.

The order matters more than any individual number here. Calibrate first, then choose on power. Choosing on power first systematically favours whichever method is most willing to reject, which is the opposite of what a programme wants from the estimate it will defend to a regulator.

5. Model checking

No diagnostics were recorded. An analysis with no model checking is not therefore sound; it is unexamined.

6. Discussion

There are no findings to interpret.

7. Conclusions

Five defensible ways to estimate a heat-pump retrofit's saving, each resting on different assumptions. Which of them actually holds its size on the panel this programme will really have?

No finding was recorded, so no conclusion follows.

8. Limitations

No assumption in the record is left unresolved. That is a statement about the record, not a guarantee about the world.

9. Provenance

Every number in this document resolves to a quantity in the evidence record below. Prose was checked against that record: any numeral not traceable to it, and any claim the record does not itself make, is reported rather than published.

Field	Value
field	Energy
source	examples/ walkthrough record
steps	5
evidence hash	97df4b07c1055e034210bb69df2b899c30aa419e2b24bf5fd80e11b95c17ba9d

Table 2. Run